

**Final Basis of Design Report
Water
For
Finish Line Auto Club – Perimeter Center
Princess Drive and Loop 101
Scottsdale, Arizona 85255**

**COS Plan Check #: 1059-PA-2024
COS Case #: 19-ZN-2004#2**



August 2025

**Prepared by:
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H.E. PROJECT NO. LGEC332

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1.0 Introduction

This design report has been prepared under a contract from LGE Corporation, developer of the Finish Line Auto Club project. The purpose of this report is to provide a water analysis, required by the City of Scottsdale, to support this development. Preparation of this report has been done according to the procedures detailed in Chapter 6 of the City of Scottsdale's *Design Standards and Policy Manual* (Reference 1).

This development project is located along the west side of N. Perimeter Drive and along the east side of N. Pacesetter Way, between E. Pacesetter Way and E. Princess Drive, within the City of Scottsdale, Maricopa County, Arizona. The site is specifically located in the south half of Section 36, Township 4 North, Range 4 East of the Gila and Salt River Base and Meridian. Figure 1, in Appendix A, illustrates the location of the project site in relation to the City of Scottsdale street system. Access to the site is provided from N. Perimeter Drive and N. Pacesetter Way.

The proposed project site is currently a developed parcel within the Perimeter Center master development. The existing parcel is bound by existing commercial developments to the north and south, N. Perimeter Drive to the east and N. Pacesetter Way to the west.

The development proposes the construction of five new commercial buildings and one new industrial building. Site improvements will include construction of driveway entrances, parking lots, sidewalk/hardscape, landscape areas, and supporting infrastructure including new storm water drainage systems, water, sewer and fire line services. The overall project site is approximately 10.394± acres.

2.0 Existing Condition

The existing site consists of a single commercial building and associated parking lots with parking canopies. The site is located in Pressure Zone 4. Adjacent to the site, there is an existing 8-inch DIP public water main along the east side of N. Pacesetter Way and an existing 12-inch DIP public water main along the east side of N. Perimeter Drive. There is an existing 8-inch public water line, enveloped by a 12-foot existing public water line easement, that runs through the site with a connection to the aforementioned existing public main in N. Perimeter Drive and extends to an existing 8-inch public main in E. Princess Drive. This existing 8-inch public water line currently services the existing building on the site and the existing building south of the site. There are also two existing fire hydrants connected to this existing 8-inch public water line.

3.0 Proposed Water System

Two new public 8-inch water lines will be looped through the site within 20-foot public water line easements and will provide fire protection along with domestic and landscape services. One of the water line loops will service the proposed industrial building and the other will service the five proposed commercial buildings. The industrial building water line loop will have two connections to the existing 8-inch DIP public water main along the east side of N. Pacesetter Way. The water line loop servicing the

commercial buildings will have two connections to the existing 8-inch public water line that runs through the site within a 12-foot public water line easement. A portion of the existing onsite public water line and easement, along with one existing fire hydrant, will be abandoned and removed to accommodate the proposed site plan. Also, an existing fire hydrant along N. Pacesetter Way will be removed and relocated to accommodate for the proposed northwest driveway entrance. Approved backflow devices will be located at each connection to the public water lines. Respective domestic, landscape, and fire services will be taken from the new public waterline loops. Additionally, there are ten proposed onsite fire hydrants. The Conceptual Utility Plan, in Appendix A, depicts the proposed water improvements.

4.0 Projected System Demands

The Average Day Demand, Peak Hour Demand, and Maximum Day Demand for this development were derived using unit flow requirements, for industrial and commercial land uses, from the City of Scottsdale Design Standards & Policies Manual for Water, Figure 6.1-2, see Appendix C. Per this figure, the average day water demand is 1.44 gallons per minute (GPM) per acre for industrial, and 0.00111 GPM per square foot for commercial. The site is currently a single parcel that will be divided between the proposed industrial building and the five proposed commercial buildings. The proposed parcel line will be used to determine the site acreage for the industrial building domestic demands. The complete water demand calculations for the proposed buildings can be found in Appendix B. The Maximum Day Demand is 2.0 times the Average Day Demand, and the Peak Hour Demand is 3.5 times the Average Day Demand.

The proposed industrial building is 51,866 square feet and will be type V-B construction. The largest proposed commercial building is Building A. Building A is 63,020 square feet and will also be type V-B construction. Per the 2021 International Fire Code, the minimum base fire flow rate for buildings of this size and construction type is 6,250 gallons per minute (GPM) and 6,750 GPM, respectively. Because these building will be protected by an approved sprinkler system, per NFPA 13, the required fire flow may be 25% of the table value, but not below 1,500 GPM. Fire flows of 1,563 GPM for the industrial building and 1,688 GPM for the largest proposed commercial building are used for the site analysis. Per City of Scottsdale, pressure requirements, minimum acceptable design pressures are 30 psi at the hydrant under design fire flow requirements and 15 psi at all domestic demand nodes, and minimum residual pressure, while applying initial service line design flow, of 50 psi at highest finished floor for domestic demand.

5.0 Water Model

A fire flow test was completed on April 14th, 2025 by Arizona Flow Testing, LLC and the results are shown in Appendix D. This test data was used to model the proposed system using WaterCad, a pipe network analysis program by Haestad Methods. A reservoir and pump were added to the model near the flow test location to simulate pressure versus flow curve. The hydrant test was adjusted per the City of Scottsdale requirements. Note that the pipes PX-1 and PX-2 connecting the pump and reservoir are

not a part of the system and are oversized to 120-inches to minimize system losses. Pipes and junctions were added to the network model matching the pipe sizes, materials, and elevations of the proposed and existing system.

The fire flow model was set up such that 1,563 GPM and 1,688 GPM, for the proposed industrial building and Building A, respectively, is taken from the hydrant furthest from the proposed connections to the existing public system. The proposed industrial building and commercial buildings were analyzed as separate scenarios. The lowest resultant pressure based upon the required fire flow and the max day demand was calculated to be 60.01 psi for the industrial building scenario and 59.46 psi for the commercial building scenario. This is above the 30-psi fire flow minimum pressure. The resultant pressure for the peak flow is 67.48 psi for the industrial building scenario and 67.01 psi for the commercial building scenario, which are greater than the minimum peak flow pressure of 50 psi. Results and data from the WaterCAD are shown in Appendix B.

6.0 Conclusion

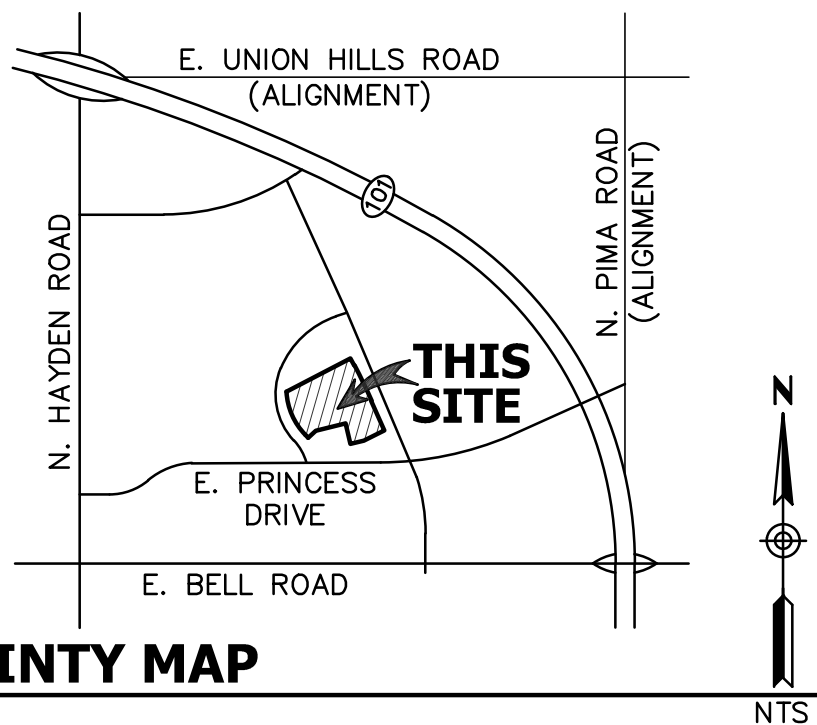
Based on the results of this study, it can be concluded that the existing and proposed water systems are adequate to support this development.

7.0 References

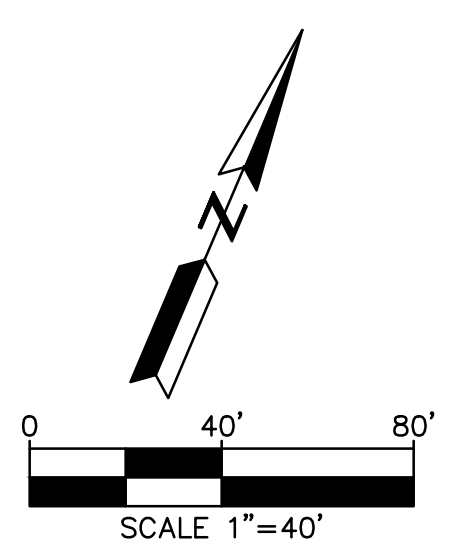
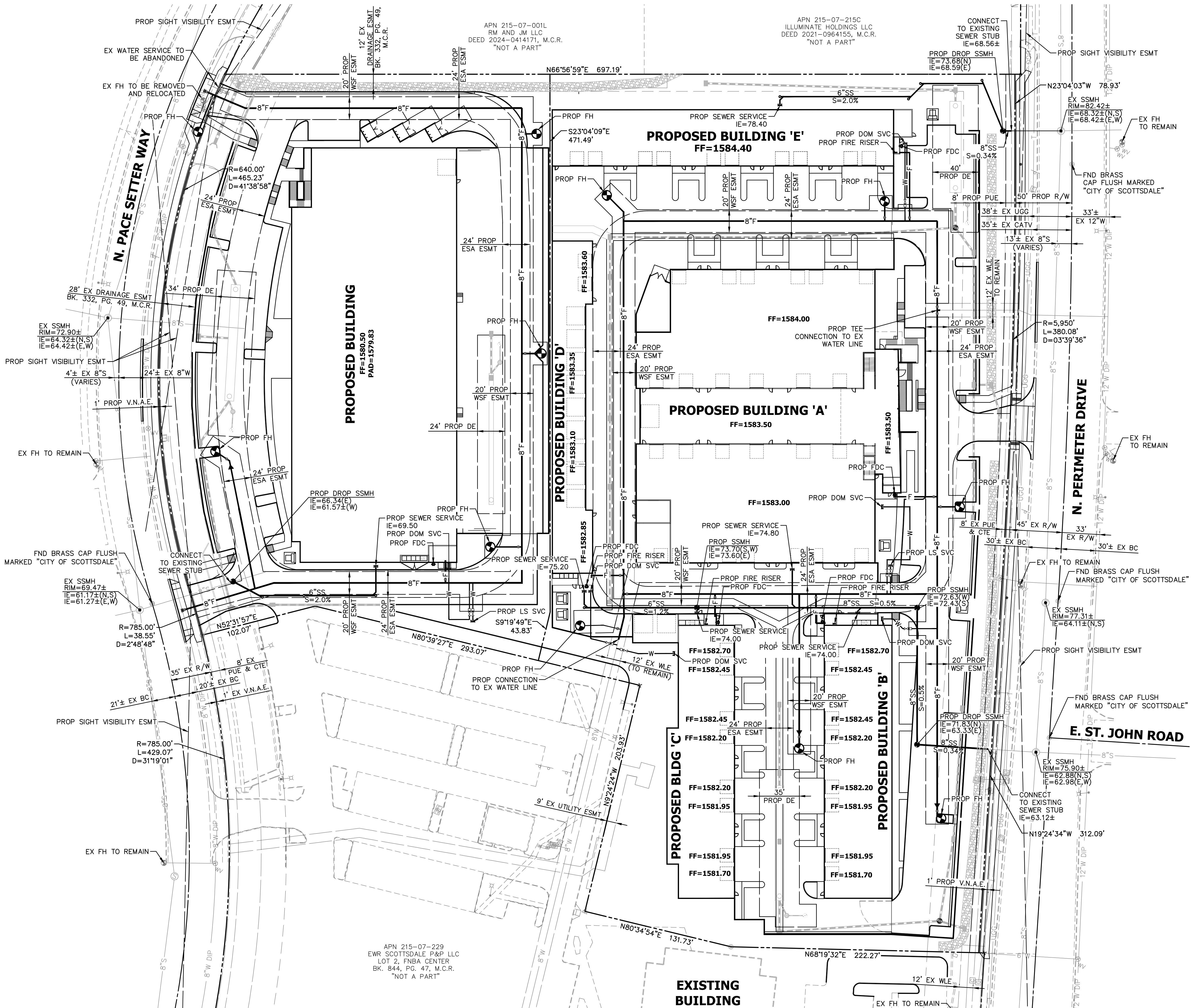
- 1) City of Scottsdale Design Standard & Policies Manual, January 2018.
- 2) International Code Council International Fire Code (2021).

Appendix A

Vicinity Map and Conceptual Utility Plan



**VICINITY MAP
FIGURE 1**



NO.	DATE	REVISION	BY

DESIGN BY: BKS
DRAWN BY: MM
CHECKED BY: BKS

HUNTER
ENGINEERING
10446 NORTH 74TH STREET
SUITE 140
SCOTTSDALE, AZ 85258
P 480 991 3985



**CONCEPTUAL UTILITY PLAN
FOR
FINISH LINE AUTO CLUB
PRINCESS DRIVE AND LOOP 101
SCOTTSDALE, ARIZONA**

CONTACT ARIZONA 811 AT LEAST 2 FULL
WORKING DAYS BEFORE YOU BEGIN EXCAVATION
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THESE PLANS ARE
NOT APPROVED FOR
CONSTRUCTION
WITHOUT AN
APPROVED SIGNATURE
FROM THE GOVERNING
MUNICIPALITY.

PROJECT NAME:
FINISH LINE
AUTO
CLUB

HE NO.: LGEC332
SCALE: 1"=40'

SHEET:
C4

Appendix B
Design Data and Calculations

WATER FLOW TEST REPORT

Project: Finish Line - Perimeter Center
Project Number: LGEC332

TOTAL FLOW DURING TEST: 2946 GPM
STATIC READING: 72 PSI RESIDUAL: 60 PSI
RESULTS: AT 20 PSI RESIDUAL 6503 GPM AT 0 PSI 7752 GPM
REMARKS:

	Shutoff Head	Design	Max Operating
	72 psi	60 psi	0 psi
psi X 2.31 = ft	166.32 ft	138.6 ft	0 ft
		2946 flow	7752 flow

Project: Finish Line Auto Club - Perimeter Center
Project Number: LGEC332
City: Scottsdale
Date: 7/17/2025

SITE DOMESTIC DEMAND SUMMARY

Finish Line Auto Club (Perimeter Drive)	Site Use	Building Area (sf)	Average Day Demand (gpm) Gross Bldg Area (sf) per Table 6-1.2 Average Day Water Demands	Average Day Demand (ADD) (gpm)	Max Day Demand (MDD) (ADD*2.0) (gpm)	Peak Hour Demand (PHD) (ADD*3.5) (gpm)
Building A	Commercial	63,020	0.001110 gpm per sf	69.95	139.90	244.83
Building B	Commercial	17,877	0.001110 gpm per sf	19.84	39.69	69.45
Building C	Commercial	17,337	0.001110 gpm per sf	19.24	38.49	67.35
Building D	Commercial	8,712	0.001110 gpm per sf	9.67	19.34	33.85
Building E	Commercial	18,558	0.001110 gpm per sf	20.60	41.20	72.10
Subtotal		125,504		139.31	278.62	487.58

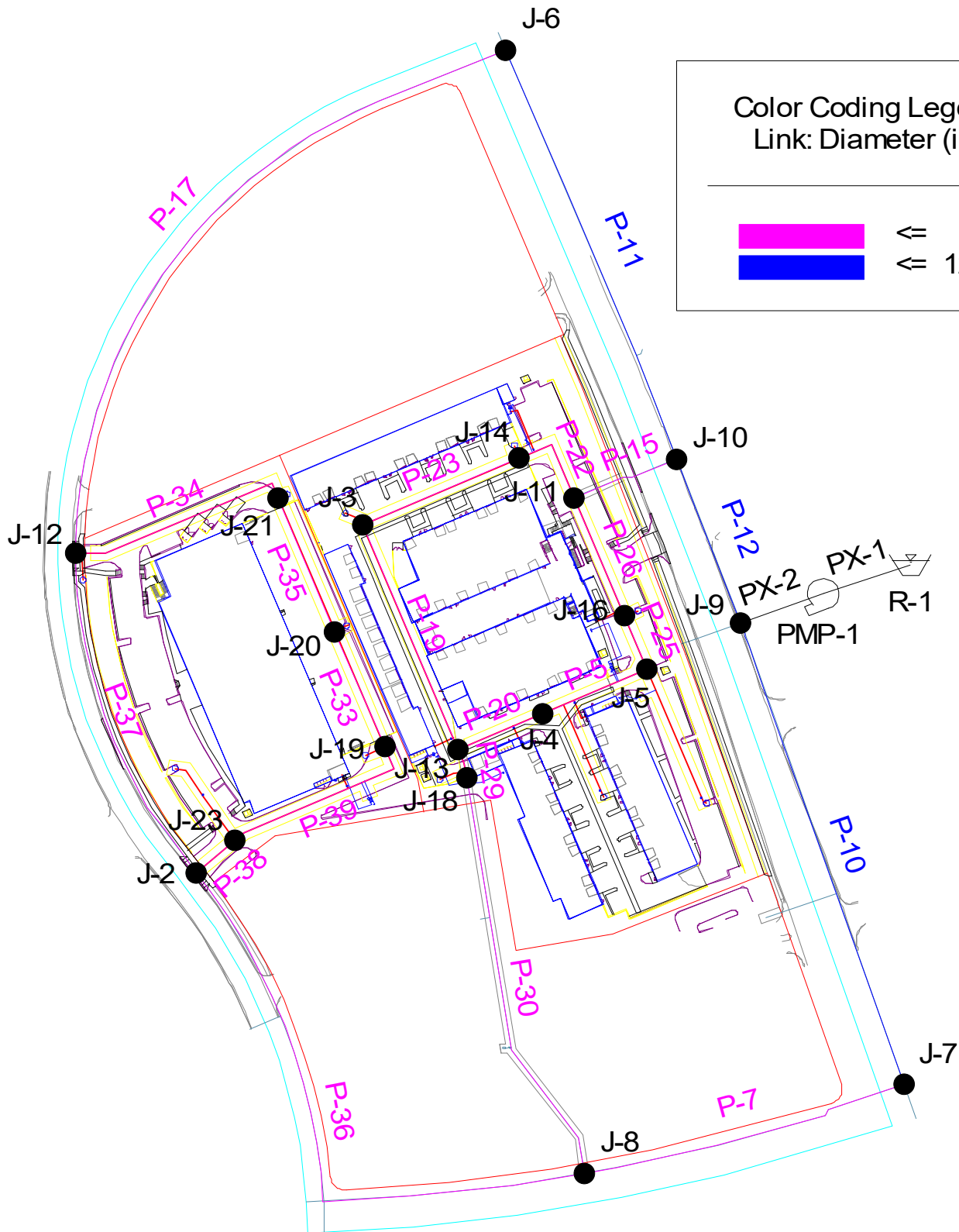
Industrial Building (Pacesetter Way)	Site Use	Site Area (ac)	Average Day Demand (gpm) Site Area (ac) per Table 6-1.2 Average Day Water Demands	Average Day Demand (ADD) (gpm)	Max Day Demand (MDD) (ADD*2.0) (gpm)	Peak Hour Demand (PHD) (ADD*3.5) (gpm)
Industrial Building	Industrial	3.65	1.44 gpm per ac	5.26	10.51	18.40
Subtotal		3.65		5.26	10.51	18.40

Grand Total				144.57	289.13	505.98
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SITE FIRE FLOW SUMMARY

Building	Construction Type	Building Area (sf)	Minimum Required Fire Flow for Buildings Table B105.1 International Fire Code	Min Fire Flow 25% of Table Value (gpm) or Min 1,500 gpm
Building A	V-B	63,020	6,750 gpm	1,688
Building B	V-B	17,877	3,500 gpm	1,500
Building C	V-B	17,337	3,500 gpm	1,500
Building D	V-B	8,712	2,500 gpm	1,500
Building E	V-B	18,558	3,750 gpm	1,500
Industrial Building	V-B	51,866	6,250 gpm	1,563

Scenario: Finish Line - Perimeter Center



Color Coding Legend
Link: Diameter (in)

	≤ 8.0
	≤ 12.0

Scenario: Average Day - Industrial

Steady State Analysis

Junction Report

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,744.07	69.45
J-4	1,582.57	0.00	1,744.07	69.87
J-5	1,581.75	0.00	1,744.07	70.23
J-6	1,588.10	0.00	1,744.07	67.48
J-7	1,569.80	0.00	1,744.07	75.40
J-8	1,567.50	0.00	1,744.07	76.39
J-9	1,577.75	0.00	1,744.07	71.96
J-10	1,580.60	0.00	1,744.07	70.73
J-11	1,581.50	0.00	1,744.07	70.34
J-12	1,575.33	0.00	1,744.07	73.01
J-13	1,582.70	0.00	1,744.07	69.82
J-16	1,581.20	0.00	1,744.07	70.47
J-18	1,579.00	0.00	1,744.07	71.42
J-19	1,575.40	5.26	1,744.07	72.98
J-20	1,579.00	0.00	1,744.07	71.42
J-21	1,581.40	0.00	1,744.07	70.38
J-2	1,569.25	0.00	1,744.07	75.64
J-14	1,583.77	0.00	1,744.07	69.35
J-23	1,571.45	0.00	1,744.07	74.68

Scenario: Average Day - Industrial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	-0.56	1,744.07	1,744.07	0.00	0.00	0.00
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	1.62	1,744.07	1,744.07	0.00	0.00	0.01
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	1.62	1,744.07	1,744.07	0.00	0.00	0.00
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	5.26	1,577.75	1,577.75	0.00	0.00	0.00
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	5.26	1,744.07	1,744.07	0.00	0.00	0.00
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-2.58	1,744.07	1,744.07	0.00	0.00	0.01
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-3.64	1,744.07	1,744.07	0.00	0.00	0.01
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-1.05	1,744.07	1,744.07	0.00	0.00	0.01
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-2.58	1,744.07	1,744.07	0.00	0.00	0.02
P-19	J-3	J-13	327.00	8.0	PVC	150.0	0.49	1,744.07	1,744.07	0.00	0.00	0.00
P-20	J-13	J-4	123.00	8.0	PVC	150.0	-0.56	1,744.07	1,744.07	0.00	0.00	0.00
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-0.56	1,744.07	1,744.07	0.00	0.00	0.00
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-0.56	1,744.07	1,744.07	0.00	0.00	0.00
P-29	J-13	J-18	41.00	8.0	PVC	150.0	1.05	1,744.07	1,744.07	0.00	0.00	0.01
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	1.05	1,744.07	1,744.07	0.00	0.00	0.01
P-34	J-12	J-21	304.00	8.0	PVC	150.0	2.17	1,744.07	1,744.07	0.00	0.00	0.01
P-33	J-20	J-19	168.00	8.0	PVC	150.0	2.17	1,744.07	1,744.07	0.00	0.00	0.01
P-35	J-21	J-20	194.00	8.0	PVC	150.0	2.17	1,744.07	1,744.07	0.00	0.00	0.01
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	2.68	1,744.07	1,744.07	0.00	0.00	0.02
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-0.41	1,744.07	1,744.07	0.00	0.00	0.00
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-0.49	1,744.07	1,744.07	0.00	0.00	0.00
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-0.49	1,744.07	1,744.07	0.00	0.00	0.00
P-38	J-2	J-23	67.00	8.0	PVC	150.0	3.09	1,744.07	1,744.07	0.00	0.00	0.02
P-39	J-23	J-19	266.00	8.0	PVC	150.0	3.09	1,744.07	1,744.07	0.00	0.00	0.02

Scenario: Max Day - Industrial

Steady State Analysis

Junction Report

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,744.07	69.45
J-4	1,582.57	0.00	1,744.07	69.87
J-5	1,581.75	0.00	1,744.07	70.23
J-6	1,588.10	0.00	1,744.07	67.48
J-7	1,569.80	0.00	1,744.07	75.40
J-8	1,567.50	0.00	1,744.07	76.39
J-9	1,577.75	0.00	1,744.07	71.96
J-10	1,580.60	0.00	1,744.07	70.73
J-11	1,581.50	0.00	1,744.07	70.34
J-12	1,575.33	0.00	1,744.07	73.00
J-13	1,582.70	0.00	1,744.07	69.82
J-16	1,581.20	0.00	1,744.07	70.47
J-18	1,579.00	0.00	1,744.07	71.42
J-19	1,575.40	10.51	1,744.07	72.97
J-20	1,579.00	0.00	1,744.07	71.42
J-21	1,581.40	0.00	1,744.07	70.38
J-2	1,569.25	0.00	1,744.07	75.64
J-14	1,583.77	0.00	1,744.07	69.35
J-23	1,571.45	0.00	1,744.07	74.68

Scenario: Max Day - Industrial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	-1.13	1,744.07	1,744.07	0.00	0.00	0.01
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	3.24	1,744.07	1,744.07	0.00	0.00	0.02
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	3.24	1,744.07	1,744.07	0.00	0.00	0.01
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	10.51	1,577.75	1,577.75	0.00	0.00	0.00
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	10.51	1,744.07	1,744.07	0.00	0.00	0.00
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-5.16	1,744.07	1,744.07	0.00	0.00	0.01
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-7.27	1,744.07	1,744.07	0.00	0.00	0.02
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-2.11	1,744.07	1,744.07	0.00	0.00	0.01
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-5.16	1,744.07	1,744.07	0.00	0.00	0.03
P-19	J-3	J-13	327.00	8.0	PVC	150.0	0.98	1,744.07	1,744.07	0.00	0.00	0.01
P-20	J-13	J-4	123.00	8.0	PVC	150.0	-1.13	1,744.07	1,744.07	0.00	0.00	0.01
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-1.13	1,744.07	1,744.07	0.00	0.00	0.01
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-1.13	1,744.07	1,744.07	0.00	0.00	0.01
P-29	J-13	J-18	41.00	8.0	PVC	150.0	2.11	1,744.07	1,744.07	0.00	0.00	0.01
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	2.11	1,744.07	1,744.07	0.00	0.00	0.01
P-34	J-12	J-21	304.00	8.0	PVC	150.0	4.34	1,744.07	1,744.07	0.00	0.00	0.03
P-33	J-20	J-19	168.00	8.0	PVC	150.0	4.34	1,744.07	1,744.07	0.00	0.00	0.03
P-35	J-21	J-20	194.00	8.0	PVC	150.0	4.34	1,744.07	1,744.07	0.00	0.00	0.03
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	5.35	1,744.07	1,744.07	0.00	0.00	0.03
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-0.82	1,744.07	1,744.07	0.00	0.00	0.01
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-0.98	1,744.07	1,744.07	0.00	0.00	0.01
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-0.98	1,744.07	1,744.07	0.00	0.00	0.01
P-38	J-2	J-23	67.00	8.0	PVC	150.0	6.17	1,744.07	1,744.07	0.00	0.00	0.04
P-39	J-23	J-19	266.00	8.0	PVC	150.0	6.17	1,744.07	1,744.07	0.00	0.00	0.04

Scenario: Peak Hour - Industrial

Steady State Analysis

Junction Report

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,744.07	69.45
J-4	1,582.57	0.00	1,744.07	69.87
J-5	1,581.75	0.00	1,744.07	70.23
J-6	1,588.10	0.00	1,744.07	67.48
J-7	1,569.80	0.00	1,744.07	75.40
J-8	1,567.50	0.00	1,744.07	76.39
J-9	1,577.75	0.00	1,744.07	71.96
J-10	1,580.60	0.00	1,744.07	70.72
J-11	1,581.50	0.00	1,744.07	70.34
J-12	1,575.33	0.00	1,744.06	73.00
J-13	1,582.70	0.00	1,744.07	69.82
J-16	1,581.20	0.00	1,744.07	70.47
J-18	1,579.00	0.00	1,744.07	71.42
J-19	1,575.40	18.40	1,744.06	72.97
J-20	1,579.00	0.00	1,744.06	71.42
J-21	1,581.40	0.00	1,744.06	70.38
J-2	1,569.25	0.00	1,744.06	75.63
J-14	1,583.77	0.00	1,744.07	69.35
J-23	1,571.45	0.00	1,744.06	74.68

Scenario: Peak Hour - Industrial

Steady State Analysis

Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	-1.97	1,744.07	1,744.07	0.00	0.00	0.01
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	5.68	1,744.07	1,744.07	0.00	0.00	0.04
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	5.68	1,744.07	1,744.07	0.00	0.00	0.02
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	18.40	1,577.75	1,577.75	0.00	0.00	0.00
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	18.40	1,744.07	1,744.07	0.00	0.00	0.00
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-9.03	1,744.07	1,744.07	0.00	0.00	0.03
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-12.72	1,744.07	1,744.07	0.00	0.00	0.04
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-3.69	1,744.07	1,744.07	0.00	0.00	0.02
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-9.03	1,744.06	1,744.07	0.00	0.00	0.06
P-19	J-3	J-13	327.00	8.0	PVC	150.0	1.71	1,744.07	1,744.07	0.00	0.00	0.01
P-20	J-13	J-4	123.00	8.0	PVC	150.0	-1.97	1,744.07	1,744.07	0.00	0.00	0.01
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-1.97	1,744.07	1,744.07	0.00	0.00	0.01
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-1.97	1,744.07	1,744.07	0.00	0.00	0.01
P-29	J-13	J-18	41.00	8.0	PVC	150.0	3.69	1,744.07	1,744.07	0.00	0.00	0.02
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	3.69	1,744.07	1,744.07	0.00	0.00	0.02
P-34	J-12	J-21	304.00	8.0	PVC	150.0	7.60	1,744.06	1,744.06	0.00	0.00	0.05
P-33	J-20	J-19	168.00	8.0	PVC	150.0	7.60	1,744.06	1,744.06	0.00	0.00	0.05
P-35	J-21	J-20	194.00	8.0	PVC	150.0	7.60	1,744.06	1,744.06	0.00	0.00	0.05
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	9.37	1,744.07	1,744.06	0.00	0.00	0.06
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-1.43	1,744.06	1,744.06	0.00	0.00	0.01
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-1.71	1,744.07	1,744.07	0.00	0.00	0.01
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-1.71	1,744.07	1,744.07	0.00	0.00	0.01
P-38	J-2	J-23	67.00	8.0	PVC	150.0	10.80	1,744.06	1,744.06	0.00	0.00	0.07
P-39	J-23	J-19	266.00	8.0	PVC	150.0	10.80	1,744.06	1,744.06	0.00	0.00	0.07

Scenario: Fire+Max Day - Industrial

Steady State Analysis

Junction Report

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,734.24	65.19
J-4	1,582.57	0.00	1,734.17	65.59
J-5	1,581.75	0.00	1,734.25	65.98
J-6	1,588.10	0.00	1,733.75	63.02
J-7	1,569.80	0.00	1,734.96	71.46
J-8	1,567.50	0.00	1,732.83	71.53
J-9	1,577.75	0.00	1,735.39	68.20
J-10	1,580.60	0.00	1,734.69	66.67
J-11	1,581.50	0.00	1,734.38	66.14
J-12	1,575.33	0.00	1,722.83	63.82
J-13	1,582.70	0.00	1,734.11	65.51
J-16	1,581.20	0.00	1,734.29	66.23
J-18	1,579.00	0.00	1,734.04	67.08
J-19	1,575.40	10.51	1,719.83	62.49
J-20	1,579.00	1,563.00	1,718.36	60.29
J-21	1,581.40	0.00	1,720.10	60.01
J-2	1,569.25	0.00	1,722.83	66.45
J-14	1,583.77	0.00	1,734.33	65.14
J-23	1,571.45	0.00	1,722.23	65.23

Scenario: Fire+Max Day - Industrial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	-167.83	1,734.17	1,734.25	0.08	0.51	1.07
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	484.20	1,734.96	1,732.83	2.13	4.77	3.09
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	484.20	1,735.39	1,734.96	0.43	0.66	1.37
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	1,573.51	1,577.75	1,577.75	0.00	0.00	0.04
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	1,573.51	1,735.39	1,735.39	0.00	0.00	0.04
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-775.68	1,733.75	1,734.69	0.94	1.58	2.20
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-1,089.31	1,734.69	1,735.39	0.70	2.97	3.09
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-313.63	1,734.38	1,734.69	0.31	2.13	2.00
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-775.68	1,722.83	1,733.75	10.92	11.41	4.95
P-19	J-3	J-13	327.00	8.0	PVC	150.0	145.80	1,734.24	1,734.11	0.13	0.40	0.93
P-20	J-13	J-4	123.00	8.0	PVC	150.0	-167.83	1,734.11	1,734.17	0.06	0.51	1.07
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-167.83	1,734.25	1,734.29	0.04	0.51	1.07
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-167.83	1,734.29	1,734.38	0.09	0.51	1.07
P-29	J-13	J-18	41.00	8.0	PVC	150.0	313.63	1,734.11	1,734.04	0.07	1.64	2.00
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	313.63	1,734.04	1,732.83	1.21	2.13	2.00
P-34	J-12	J-21	304.00	8.0	PVC	150.0	786.19	1,722.83	1,720.10	2.73	8.98	5.02
P-33	J-20	J-19	168.00	8.0	PVC	150.0	-776.81	1,718.36	1,719.83	1.47	8.78	4.96
P-35	J-21	J-20	194.00	8.0	PVC	150.0	786.19	1,720.10	1,718.36	1.74	8.98	5.02
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	797.83	1,732.83	1,722.83	10.00	12.02	5.09
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	10.51	1,722.83	1,722.83	0.00	0.00	0.07
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-145.80	1,734.24	1,734.33	0.09	0.40	0.93
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-145.80	1,734.33	1,734.38	0.05	0.40	0.93
P-38	J-2	J-23	67.00	8.0	PVC	150.0	787.32	1,722.83	1,722.23	0.60	9.00	5.03
P-39	J-23	J-19	266.00	8.0	PVC	150.0	787.32	1,722.23	1,719.83	2.39	9.00	5.03

Scenario: Average Day - Commercial **Steady State Analysis** **Junction Report**

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,743.92	69.39
J-4	1,582.57	0.00	1,743.92	69.81
J-5	1,581.75	89.80	1,743.91	70.16
J-6	1,588.10	0.00	1,743.96	67.43
J-7	1,569.80	0.00	1,743.97	75.35
J-8	1,567.50	0.00	1,743.95	76.34
J-9	1,577.75	0.00	1,743.97	71.92
J-10	1,580.60	0.00	1,743.96	70.68
J-11	1,581.50	0.00	1,743.93	70.28
J-12	1,575.33	0.00	1,743.96	72.96
J-13	1,582.70	0.00	1,743.92	69.75
J-16	1,581.20	0.00	1,743.92	70.40
J-18	1,579.00	28.91	1,743.92	71.35
J-19	1,575.40	0.00	1,743.96	72.93
J-20	1,579.00	0.00	1,743.96	71.37
J-21	1,581.40	0.00	1,743.96	70.33
J-2	1,569.25	0.00	1,743.96	75.59
J-14	1,583.77	20.60	1,743.93	69.29
J-23	1,571.45	0.00	1,743.96	74.64

Scenario: Average Day - Commercial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	34.42	1,743.92	1,743.91	0.00	0.03	0.22
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	33.83	1,743.97	1,743.95	0.02	0.03	0.22
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	33.83	1,743.97	1,743.97	0.00	0.00	0.10
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	139.31	1,577.75	1,577.75	0.00	0.00	0.00
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	139.31	1,743.97	1,743.97	0.00	0.00	0.00
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-11.29	1,743.96	1,743.96	0.00	0.00	0.03
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-105.48	1,743.96	1,743.97	0.01	0.04	0.30
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-94.19	1,743.93	1,743.96	0.03	0.23	0.60
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-11.29	1,743.96	1,743.96	0.00	0.00	0.07
P-19	J-3	J-13	327.00	8.0	PVC	150.0	18.21	1,743.92	1,743.92	0.00	0.01	0.12
P-20	J-13	J-4	123.00	8.0	PVC	150.0	34.42	1,743.92	1,743.92	0.00	0.03	0.22
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-55.38	1,743.91	1,743.92	0.01	0.06	0.35
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-55.38	1,743.92	1,743.93	0.01	0.07	0.35
P-29	J-13	J-18	41.00	8.0	PVC	150.0	-16.21	1,743.92	1,743.92	0.00	0.01	0.10
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	-45.12	1,743.92	1,743.95	0.03	0.06	0.29
P-34	J-12	J-21	304.00	8.0	PVC	150.0	4.89	1,743.96	1,743.96	0.00	0.00	0.03
P-33	J-20	J-19	168.00	8.0	PVC	150.0	4.89	1,743.96	1,743.96	0.00	0.00	0.03
P-35	J-21	J-20	194.00	8.0	PVC	150.0	4.89	1,743.96	1,743.96	0.00	0.00	0.03
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	-11.29	1,743.95	1,743.96	0.00	0.00	0.07
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-6.40	1,743.96	1,743.96	0.00	0.00	0.04
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-18.21	1,743.92	1,743.93	0.00	0.01	0.12
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-38.81	1,743.93	1,743.93	0.00	0.03	0.25
P-38	J-2	J-23	67.00	8.0	PVC	150.0	-4.89	1,743.96	1,743.96	0.00	0.00	0.03
P-39	J-23	J-19	266.00	8.0	PVC	150.0	-4.89	1,743.96	1,743.96	0.00	0.00	0.03

Scenario: Max Day - Commercial

Steady State Analysis

Junction Report

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,743.54	69.22
J-4	1,582.57	0.00	1,743.52	69.63
J-5	1,581.75	179.59	1,743.50	69.98
J-6	1,588.10	0.00	1,743.68	67.31
J-7	1,569.80	0.00	1,743.71	75.24
J-8	1,567.50	0.00	1,743.65	76.21
J-9	1,577.75	0.00	1,743.72	71.81
J-10	1,580.60	0.00	1,743.68	70.56
J-11	1,581.50	0.00	1,743.56	70.12
J-12	1,575.33	0.00	1,743.67	72.83
J-13	1,582.70	0.00	1,743.53	69.58
J-16	1,581.20	0.00	1,743.52	70.23
J-18	1,579.00	57.83	1,743.53	71.19
J-19	1,575.40	0.00	1,743.67	72.80
J-20	1,579.00	0.00	1,743.67	71.24
J-21	1,581.40	0.00	1,743.67	70.21
J-2	1,569.25	0.00	1,743.67	75.46
J-14	1,583.77	41.20	1,743.55	69.13
J-23	1,571.45	0.00	1,743.67	74.51

Scenario: Max Day - Commercial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	68.83	1,743.52	1,743.50	0.01	0.10	0.44
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	67.65	1,743.71	1,743.65	0.06	0.12	0.43
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	67.65	1,743.72	1,743.71	0.01	0.02	0.19
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	278.62	1,577.75	1,577.75	0.00	0.00	0.01
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	278.62	1,743.72	1,743.72	0.00	0.00	0.01
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-22.58	1,743.68	1,743.68	0.00	0.00	0.06
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-210.97	1,743.68	1,743.72	0.03	0.14	0.60
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-188.39	1,743.56	1,743.68	0.12	0.83	1.20
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-22.58	1,743.67	1,743.68	0.02	0.02	0.14
P-19	J-3	J-13	327.00	8.0	PVC	150.0	36.42	1,743.54	1,743.53	0.01	0.03	0.23
P-20	J-13	J-4	123.00	8.0	PVC	150.0	68.83	1,743.53	1,743.52	0.01	0.10	0.44
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-110.76	1,743.50	1,743.52	0.02	0.24	0.71
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-110.76	1,743.52	1,743.56	0.04	0.24	0.71
P-29	J-13	J-18	41.00	8.0	PVC	150.0	-32.40	1,743.53	1,743.53	0.00	0.02	0.21
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	-90.23	1,743.53	1,743.65	0.12	0.21	0.58
P-34	J-12	J-21	304.00	8.0	PVC	150.0	9.78	1,743.67	1,743.67	0.00	0.00	0.06
P-33	J-20	J-19	168.00	8.0	PVC	150.0	9.78	1,743.67	1,743.67	0.00	0.00	0.06
P-35	J-21	J-20	194.00	8.0	PVC	150.0	9.78	1,743.67	1,743.67	0.00	0.00	0.06
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	-22.58	1,743.65	1,743.67	0.01	0.02	0.14
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-12.80	1,743.67	1,743.67	0.00	0.01	0.08
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-36.42	1,743.54	1,743.55	0.01	0.03	0.23
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-77.62	1,743.55	1,743.56	0.02	0.12	0.50
P-38	J-2	J-23	67.00	8.0	PVC	150.0	-9.78	1,743.67	1,743.67	0.00	0.00	0.06
P-39	J-23	J-19	266.00	8.0	PVC	150.0	-9.78	1,743.67	1,743.67	0.00	0.00	0.06

Scenario: Peak Hour - Commercial **Steady State Analysis** **Junction Report**

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	0.00	1,742.58	68.80
J-4	1,582.57	0.00	1,742.52	69.20
J-5	1,581.75	314.28	1,742.47	69.54
J-6	1,588.10	0.00	1,742.98	67.01
J-7	1,569.80	0.00	1,743.05	74.96
J-8	1,567.50	0.00	1,742.89	75.88
J-9	1,577.75	0.00	1,743.08	71.53
J-10	1,580.60	0.00	1,742.98	70.26
J-11	1,581.50	0.00	1,742.64	69.72
J-12	1,575.33	0.00	1,742.94	72.52
J-13	1,582.70	0.00	1,742.55	69.16
J-16	1,581.20	0.00	1,742.52	69.80
J-18	1,579.00	101.20	1,742.55	70.76
J-19	1,575.40	0.00	1,742.93	72.48
J-20	1,579.00	0.00	1,742.93	70.93
J-21	1,581.40	0.00	1,742.93	69.89
J-2	1,569.25	0.00	1,742.93	75.14
J-14	1,583.77	72.10	1,742.60	68.72
J-23	1,571.45	0.00	1,742.93	74.19

Scenario: Peak Hour - Commercial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	120.45	1,742.52	1,742.47	0.04	0.28	0.77
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	118.39	1,743.05	1,742.89	0.16	0.35	0.76
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	118.39	1,743.08	1,743.05	0.03	0.05	0.34
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	487.58	1,577.75	1,577.75	0.00	0.00	0.01
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	487.58	1,743.08	1,743.08	0.00	0.00	0.01
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-39.51	1,742.98	1,742.98	0.00	0.01	0.11
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-369.19	1,742.98	1,743.08	0.09	0.40	1.05
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-329.67	1,742.64	1,742.98	0.34	2.34	2.10
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-39.51	1,742.94	1,742.98	0.04	0.05	0.25
P-19	J-3	J-13	327.00	8.0	PVC	150.0	63.74	1,742.58	1,742.55	0.03	0.09	0.41
P-20	J-13	J-4	123.00	8.0	PVC	150.0	120.45	1,742.55	1,742.52	0.03	0.28	0.77
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-193.83	1,742.47	1,742.52	0.05	0.67	1.24
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-193.83	1,742.52	1,742.64	0.12	0.67	1.24
P-29	J-13	J-18	41.00	8.0	PVC	150.0	-56.71	1,742.55	1,742.55	0.00	0.07	0.36
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	-157.91	1,742.55	1,742.89	0.34	0.60	1.01
P-34	J-12	J-21	304.00	8.0	PVC	150.0	17.11	1,742.94	1,742.93	0.00	0.01	0.11
P-33	J-20	J-19	168.00	8.0	PVC	150.0	17.11	1,742.93	1,742.93	0.00	0.01	0.11
P-35	J-21	J-20	194.00	8.0	PVC	150.0	17.11	1,742.93	1,742.93	0.00	0.01	0.11
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	-39.51	1,742.89	1,742.93	0.04	0.05	0.25
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-22.41	1,742.93	1,742.94	0.01	0.02	0.14
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-63.74	1,742.58	1,742.60	0.02	0.09	0.41
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-135.84	1,742.60	1,742.64	0.04	0.35	0.87
P-38	J-2	J-23	67.00	8.0	PVC	150.0	-17.11	1,742.93	1,742.93	0.00	0.01	0.11
P-39	J-23	J-19	266.00	8.0	PVC	150.0	-17.11	1,742.93	1,742.93	0.00	0.01	0.11

Scenario: Fire+Max Day - Commercial

Steady State Analysis

Junction Report

Label	Elevation (ft)	Base Flow (gpm)	Calculated Hydraulic Grade (ft)	Pressure (psi)
J-3	1,583.55	1,688.00	1,720.99	59.46
J-4	1,582.57	0.00	1,724.18	61.27
J-5	1,581.75	179.59	1,724.34	61.69
J-6	1,588.10	0.00	1,729.64	61.24
J-7	1,569.80	0.00	1,730.55	69.55
J-8	1,567.50	0.00	1,728.54	69.68
J-9	1,577.75	0.00	1,730.96	66.28
J-10	1,580.60	0.00	1,729.69	64.50
J-11	1,581.50	0.00	1,725.04	62.10
J-12	1,575.33	0.00	1,729.10	66.53
J-13	1,582.70	0.00	1,724.06	61.16
J-16	1,581.20	0.00	1,724.55	62.02
J-18	1,579.00	57.83	1,724.26	62.85
J-19	1,575.40	0.00	1,729.04	66.47
J-20	1,579.00	0.00	1,729.06	64.92
J-21	1,581.40	0.00	1,729.07	63.89
J-2	1,569.25	0.00	1,729.01	69.12
J-14	1,583.77	41.20	1,723.54	60.47
J-23	1,571.45	0.00	1,729.02	68.17

Scenario: Fire+Max Day - Commercial
Steady State Analysis
Pipe Report

Label	Start Node	Stop Node	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Discharge (gpm)	Upstream Structure Hydraulic Grade (ft)	Downstream Structure Hydraulic Grade (ft)	Pressure Pipe Headloss (ft)	Headloss Gradient (ft/1000ft)	Velocity (ft/s)
P-5	J-4	J-5	152.00	8.0	PVC	150.0	-241.22	1,724.18	1,724.34	0.15	1.01	1.54
P-7	J-7	J-8	446.00	8.0	Ductile Iron	130.0	469.15	1,730.55	1,728.54	2.01	4.50	2.99
P-10	J-9	J-7	654.00	12.0	Ductile Iron	130.0	469.15	1,730.96	1,730.55	0.41	0.62	1.33
PX-1	R-1	PMP-1	1.00	120.0	Ductile Iron	130.0	1,966.62	1,577.75	1,577.75	0.00	0.00	0.06
PX-2	PMP-1	J-9	1.00	120.0	Ductile Iron	130.0	1,966.62	1,730.96	1,730.96	0.00	0.00	0.06
P-11	J-6	J-10	593.00	12.0	Ductile Iron	130.0	-152.76	1,729.64	1,729.69	0.05	0.08	0.43
P-12	J-10	J-9	237.00	12.0	Ductile Iron	130.0	-1,497.47	1,729.69	1,730.96	1.27	5.35	4.25
P-15	J-11	J-10	147.00	8.0	Ductile Iron	130.0	-1,344.71	1,725.04	1,729.69	4.65	31.61	8.58
P-17	J-12	J-6	957.00	8.0	Ductile Iron	130.0	-152.76	1,729.10	1,729.64	0.54	0.56	0.98
P-19	J-3	J-13	327.00	8.0	PVC	150.0	-805.31	1,720.99	1,724.06	3.07	9.38	5.14
P-20	J-13	J-4	123.00	8.0	PVC	150.0	-241.22	1,724.06	1,724.18	0.12	1.01	1.54
P-25	J-5	J-16	77.00	8.0	PVC	150.0	-420.81	1,724.34	1,724.55	0.22	2.82	2.69
P-26	J-16	J-11	172.00	8.0	PVC	150.0	-420.81	1,724.55	1,725.04	0.49	2.82	2.69
P-29	J-13	J-18	41.00	8.0	PVC	150.0	-564.08	1,724.06	1,724.26	0.20	4.85	3.60
P-30	J-18	J-8	565.00	8.0	Ductile Iron	130.0	-621.91	1,724.26	1,728.54	4.28	7.58	3.97
P-34	J-12	J-21	304.00	8.0	PVC	150.0	66.14	1,729.10	1,729.07	0.03	0.09	0.42
P-33	J-20	J-19	168.00	8.0	PVC	150.0	66.14	1,729.06	1,729.04	0.02	0.09	0.42
P-35	J-21	J-20	194.00	8.0	PVC	150.0	66.14	1,729.07	1,729.06	0.02	0.09	0.42
P-36	J-8	J-2	832.00	8.0	Ductile Iron	130.0	-152.76	1,728.54	1,729.01	0.47	0.56	0.98
P-37	J-2	J-12	465.00	8.0	Ductile Iron	130.0	-86.62	1,729.01	1,729.10	0.09	0.20	0.55
P-23	J-3	J-14	229.00	8.0	PVC	150.0	-882.69	1,720.99	1,723.54	2.55	11.12	5.63
P-22	J-14	J-11	124.00	8.0	PVC	150.0	-923.89	1,723.54	1,725.04	1.50	12.10	5.90
P-38	J-2	J-23	67.00	8.0	PVC	150.0	-66.14	1,729.01	1,729.02	0.01	0.09	0.42
P-39	J-23	J-19	266.00	8.0	PVC	150.0	-66.14	1,729.02	1,729.04	0.02	0.09	0.42

Detailed Report for Reservoir: R-1

Scenario Summary

Scenario	Fire+Max Day - Commercial
Active Topology Alternative	Base-Active Topology
Physical Alternative	Base-Physical
Demand Alternative	Demand-Fire+Max Day
Initial Settings Alternative	Base-Initial Settings
Operational Alternative	Base-Operational
Age Alternative	Base-Age Alternative
Constituent Alternative	Base-Constituent
Trace Alternative	Base-Trace Alternative
Fire Flow Alternative	Base-Fire Flow
Capital Cost Alternative	Base-Capital Cost
Energy Cost Alternative	Base-Energy Cost
User Data Alternative	Base-User Data

Global Adjustments Summary

Demand	<None>	Roughness	<None>
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Geometric Summary

X	705,783.82 ft	Elevation	1,577.75 ft
Y	962,842.39 ft	Zone	Zone

Calculated Results Summary

Time (hr)	Calculated Hydraulic Grade (ft)	Inflow (gpm)	Outflow (gpm)
0.00	1,577.75	-1,966.62	1,966.62

Detailed Report for Pump: PMP-1

Scenario Summary

Scenario	Fire+Max Day - Commercial
Active Topology Alternative	Base-Active Topology
Physical Alternative	Base-Physical
Demand Alternative	Demand-Fire+Max Day
Initial Settings Alternative	Base-Initial Settings
Operational Alternative	Base-Operational
Age Alternative	Base-Age Alternative
Constituent Alternative	Base-Constituent
Trace Alternative	Base-Trace Alternative
Fire Flow Alternative	Base-Fire Flow
Capital Cost Alternative	Base-Capital Cost
Energy Cost Alternative	Base-Energy Cost
User Data Alternative	Base-User Data

Global Adjustments Summary

Demand	<None>	Roughness	<None>
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Geometric Summary

X	705,666.53 ft	Upstream Pipe	PX-1
Y	962,803.83 ft	Downstream Pipe	PX-2
Elevation	1,577.75 ft		

Pump Definition Summary

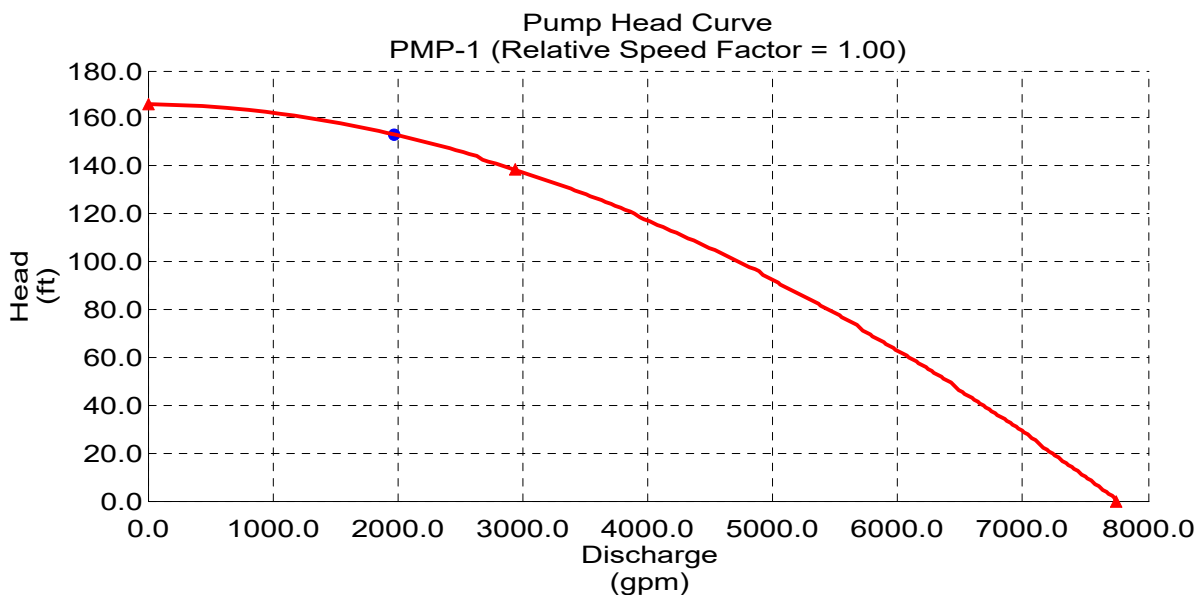
Pump Definition	Default Pump Definition
-----------------	-------------------------

Initial Status

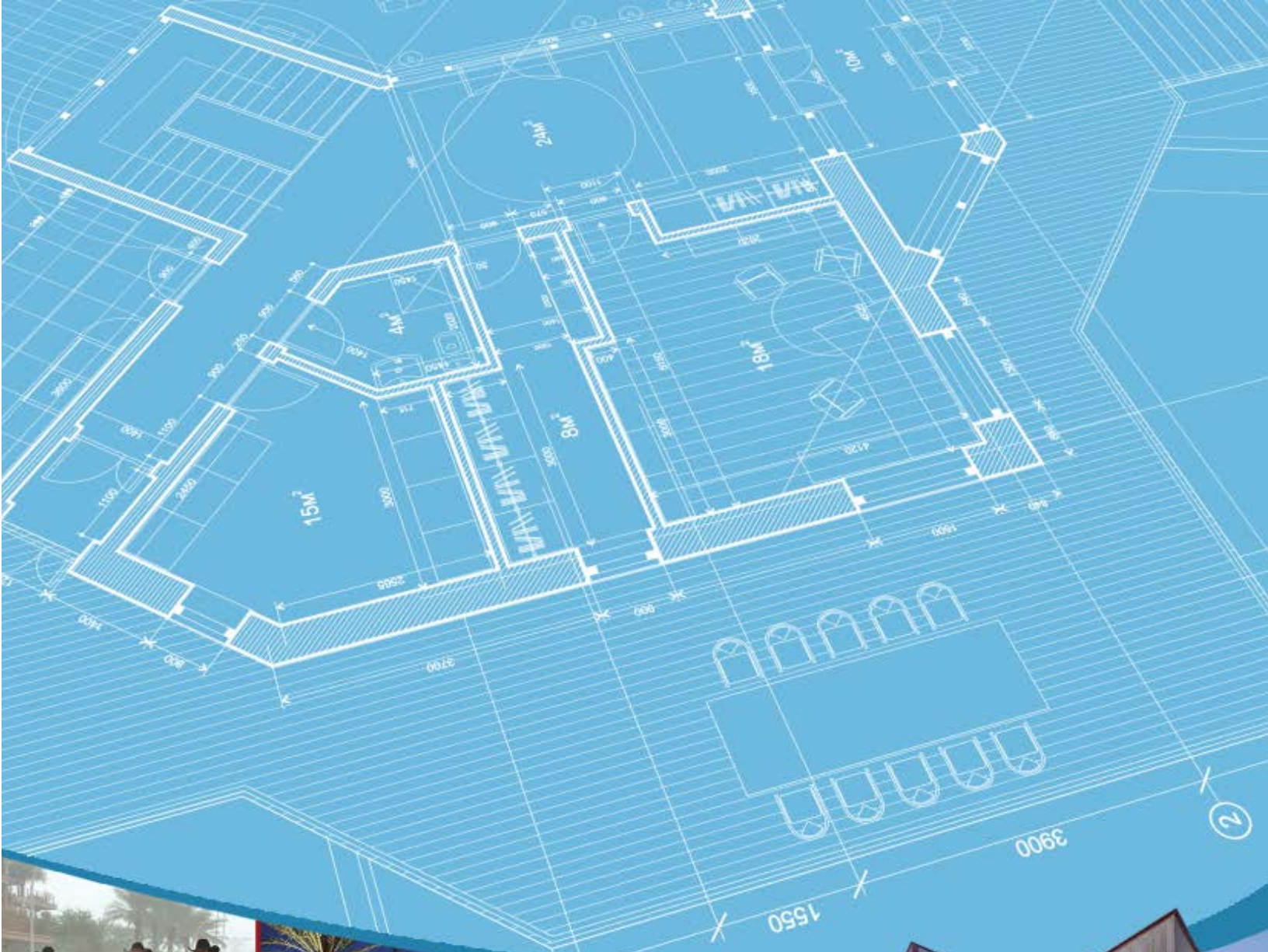
Initial Pump Status	On	Initial Relative Speed Factor	1.00
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Calculated Results Summary

Time (hr)	Control Status	Intake Pump Grade (ft)	Discharge Pump Grade (ft)	Discharge (gpm)	Pump Head (ft)	Relative Speed	Calculated Water Power (Hp)
0.00	On	1,577.75	1,730.96	1,966.62	153.21	1.00	76.07



Appendix C
Reference Information



DESIGN STANDARDS & POLICIES MANUAL

are met. The necessary process to develop the design system supply curve or grade line from the hydrant test results is described in Section 6-1.405.

3. Water demands shall generally be determined based on the unit demands in gallons per minute (gpm) listed in Figure 6-1.2, or as accepted or directed by the Water Resources Department, or as described within this chapter. The Water Resources Department reserves the right to require the use of potentially higher water demand peaking factors or directly apply demand flows for developments if deemed justified e.g. some restaurants or specialty developments.
4. Computer hydraulic modeling shall use H2ONET, WATERCAD, or EPANET software, or other Water Resource Department approved software.
5. In addition to network diagrams clearly present all inputs, details, and analysis output in organized tables.
6. Include all relevant hydraulic network diagrams listed for each of the following required hydraulic analysis scenarios:
 - a. Model Scenario 1: Average day demand in gpm at all demand nodes. (Refer to Figure 6-1.2)
 - b. Model Scenario 2: Peak hour demand in gpm at all demand nodes. (Refer to Figure 6-1.2 and Section 6-1.404 for peak hour demand)
 - c. Model Scenario 3: Maximum day demand in gpm at all demand nodes with worst case fire flow (Refer to section 6-1.501 describing fire flow determination. Refer to Figure 6-1.2 and Section 6-1.404 for maximum day demand)

Requirements:

- i. The determined fire flow must be applied to the single worst-case location in the proposed system where fire flow will be required. Typically, this is the furthest and/or highest point from the main water supply connection. If not clear what the worst-case fire flow demand point is, the fire flow shall be applied to each potential point in the model until the worst-case point is determined.
- ii. A minimum of 30 pounds per square inch (psi) must be maintained at the worst-case hydrant supply line tee/tap under this condition with a simultaneous minimum of 15 psi maintained at all domestic demand nodes (i.e., at the highest finished floor elevation and post service line and appurtenances) (Refer to scenario 4 for guidance on modeling the service line up to the demand nodes).
- d. Model Scenario 4: Maintain the minimum domestic service pressure at the worst case domestic demand node (location/elevation) under normal daily operating flow conditions termed henceforth as the *Initial Service Line Design Flow*.

Notes: A minimum of 50 psi must be maintained at the highest proposed finished floor elevation to be served, for the worst hydraulic case domestic demand node, while applying the Initial Service Line Design Flow to the node. Typically, this will be the demand node with the lowest modeled pressure in the previous scenarios. The engineer shall define the service line

details, demand node location, and determine normal daily operating hydraulic conditions/criteria as follows:

- i. Demand point location: the furthest, highest (i.e. worst case hydraulic metered node). If the service line distances and building heights are unknown, the demand node shall be located at the geometric center of the lot at elevation 12 feet above the planned finished floor elevation for single family residential. Otherwise a typical highest finished floor elevation for the development type shall be used.
- ii. Initial Service Line Design Flow: 1) Estimate the average number of water fixtures served by the subject node; 2) Use the 2015 International Plumbing Code, Appendix E, Table E103.3(2)-total load values and Table E103.3(3) to determine the normal operating flow rate; 3) Add 10 gpm minimum to the normal operating flow to account for either a hose bib or a single irrigation sprinkler zone or estimate a higher applicable intermittent or constant base flow; 4) Apply a 1.5 safety factor to the resultant total flow rate to obtain the Initial Service Line Design Flow. Note that this flow also factors into meter sizing, refer to section 6-1.416 Service Lines and Meters.
- iii. Determine the required service line and appurtenance sizing: If the sprinkler system and the domestic uses are metered through a shared meter use the greater of the resultant flow in step above, or the required fire sprinkler flow. Refer to the applicable Fire Code for sprinkler system flow and pressure requirements. Refer to section 6-1.404 Design Flows and Head Loss for design criteria on service lines.
- iv. Model pressure losses between the service tap and the demand node: Determine the losses through the water meter and the pressure reducing valve for the resultant flow and sizing from the step above. A combined 10 psi or greater loss shall be used for meter and pressure reducing valve (PRV) in scenario 4. A 5 psi or greater loss shall be used for the meter and PRV in other modeled scenarios. The service pipe friction loss portion will be per its length and diameter as included in the model.
- v. All other demand nodes in the network, other than the worst-case node shall be assigned their corresponding peak hour total use demand per Figure 6.1-2 and Section 6-1.404 during this scenario.
- vi. No fire flows are to be applied for this scenario.

H. Network Diagrams

1. Network Diagram 1: Describe the Physical Modeled Network
Refer to Sections 6-1.300, 6-1.400, and 6-1.500 and their related subsections for water network design requirements.
Present all the nodes, valves, pipes, tanks, hydrants, and pumps within the network and present the following:
 - a. Network components with IDs
 - b. Existing distributions system pipelines (label as existing) and connection to the proposed system

- c. Complete proposed water system, including:
 - d. Each/all individually metered service lines ending in a demand node
 - i. Irrigation connections with demand nodes (show both dedicated metered lines and where shared with the domestic service line post-meter)
 - ii. Fire sprinkler riser connections with demand nodes (typically dedicated connections are used for developments other than single family residential)
 - e. Elevations (ft.)
 - f. Pipe lengths (ft.)
 - g. Friction head loss coefficients used
 - h. Pipe/valve diameters (in)
 - i. Point of any changes in pipe diameter
 - j. Valve types and positions (open/closed/modulating)
 - k. Tanks: working volumes (gal), heights (ft.), diameters (ft.)
 - l. Pump curves (3-point min) or hydraulic grade line (HGL)
 - m. Note: Developments supplied from the city's water distribution system should be modeled as a pumped supply using a pump curve developed from the required hydrant test to simulate the dynamic flow vs. pressure supplied from the water distribution system. Alternatively, differing supply HGLs derived from the flow test results can be used for the different modeling scenarios.
 - n. Provide all valve, tank, pump, and system settings (levels, pressures/control set points, valve open/close/modulation rates and settings, valve coefficients, initial conditions, etc.)
 - o. Static system supply pressure (if supply is from system, develop pump curve or HGL from hydrant test)
- Shows all numerical units or provide a legend that indicates the units used
2. Network Diagram 2: Describe the Specific Demand Scenario
 - a. Label the figure with the title of specific modeling scenario
 - b. Indicate the demand (gpm) being applied to each applicable node for the scenario
 - c. Demand nodes shall include all applicable demands for the defined scenario including domestic use (indoor), irrigation (outdoor), fire sprinkler or hydrant, etc. If a node is a summation of more than one demand type describe this in a notes column in the associated tables
 - d. Provide all valve, tank, pump, or system settings that are specific to the modeling scenario
 - e. Shows all units or provide a legend that indicates the units used
 3. Network Diagram 3: Present the Scenario Analysis Results
 - Label the figure with the title of the specific modeling scenario and present the following output:
 - a. Node pressures (psi or HGL elevation)
 - b. Pipe flow (gpm)
 - c. Flow direction arrows

- d. Pipe flow velocity in feet per second (fps)
- e. Each pipe segment's head loss rate (ft. /1,000ft or psi/ft.)
- f. PRVs: Upstream and downstream pressures (psi or HGL elevation)
- g. Tanks: Inflow and outflow (gpm)
- h. Shows all units for the values presented or provide a legend on the diagram page that indicates the units used

AVERAGE DAY WATER DEMANDS ⁽¹⁾							
IN GALLONS PER DAY (GPD) ⁽²⁾				IN GALLONS PER MINUTE (GPM) ⁽²⁾⁽³⁾			
Land Use	Inside Use	Outside Use	Total Use	Inside Use	Outside Use	Total Use	Units
Residential Demand per Dwelling Unit							
< 2 dwelling unit per acre (DU/ac)	208.9	276.7	485.6	0.30	0.39	0.69	per unit
2 – 2.9 DU/ac	193.7	276.7	470.4	0.27	0.39	0.66	per unit
3 – 7.9 DU/ac	175.9	72.3	248.2	0.25	0.11	0.36	per unit
8 – 11.9 DU/ac	155.3	72.3	227.6	0.22	0.11	0.33	per unit
12 – 22 DU/ac	155.3	72.3	227.6	0.22	0.11	0.33	per unit
High Density Condominium (condo)	155.3	30	185.3	0.22	0.05	0.27	per unit
Resort Hotel (includes site amenities)	401.7	44.6	446.3	0.56	0.07	0.63	per room
Service and Employment							
Restaurant	1.2	0.1	1.3	1.67E-03	1.39E-04	1.81E-03	per square foot (sq.ft.)
Commercial/ Retail	0.7	0.1	0.8	9.73E-04	1.39E-04	1.11E-03	per sq.ft.
Commercial High Rise	0.5	0.1	0.6	6.95E-04	1.39E-04	8.34E-04	per sq.ft.

AVERAGE DAY WATER DEMANDS ⁽¹⁾							
IN GALLONS PER DAY (GPD) ⁽²⁾				IN GALLONS PER MINUTE (GPM) ⁽²⁾⁽³⁾			
Office	0.5	0.1	0.6	6.95E-04	1.39E-04	8.34E-04	per sq.ft.
Institutional	670	670	1340	0.94	0.94	1.88	per acre
Industrial	873	154	1027	1.22	0.22	1.44	per acre
Research and Development	1092	192	1284	1.52	0.27	1.79	per acre
Special Use Areas							
Natural Area Open Space	0	0	0	0.0	0.0	0.0	per acre
Developed Open Space – Parks	0	1786	1786	0.0	2.49	2.49	per acre
Developed Open Space – Golf Course	0	4285	4285	0.0	5.96	5.96	per acre
Notes: (1) These values shall not be used directly for service line or water meter sizing. (2) Gallon per day values are provided for reference only. The instantaneous gallon per minute flow rates presented are intended for use in the required hydraulic modeling scenarios. The gpm values assume a 12-hour active water use period per 24-hour day. In large or specialty developments or master plans the hydraulic analysis criteria and parameters should be discussed with the Water Resources Department. Seasonal peaking should also be considered. Upon review, the Water Resources Department reserves the right to designate flows to be used in hydraulic modeling scenarios that may be different from those presented here. (3) The hydraulic modeling peaking factors used in select modeling scenarios are to be applied to the gpm values shown here. Max day and peak hour peaking factors can be found in Section 6-1.404.							

FIGURE 6-1.2 AVERAGE DAY WATER DEMANDS

Appendix D
Fire Flow Test

Arizona Flow Testing LLC

HYDRANT FLOW TEST REPORT

Project Name:	Finish Line - Perimeter Center
Project Address:	17600 North Perimeter Center Drive, Scottsdale, Arizona, 85255
Client Project No.:	LGEC332
Arizona Flow Testing Project No.:	25290
Flow Test Permit No.:	C78455
Date and time flow test conducted:	April 14, 2025 at 6:55 AM
Data is current and reliable until:	October 14, 2025
Conducted by:	Floyd Vaughan-AZ Flow Testing, LLC (480-250-8154)
Witnessed by:	Chris Mendez – City of Scottsdale-Inspector (602-9028-9046)

Raw Test Data

Static Pressure: **90.0 PSI**
(Measured in pounds per square inch)

Residual Pressure: **78.0 PSI**
(Measured in pounds per square inch)

Pitot Pressure: **47.0 PSI**
(Measured in pounds per square inch)

Diffuser Orifice Diameter: One 4-inch Pollard Diffuser
(Measured in inches)

Coefficient of Diffuser: 0.9

Flowing GPM: **2,946 GPM**
(Measured in gallons per minute)

GPM @ 20 PSI: **7,635 GPM**

Data with 18 PSI Safety Factor

Static Pressure: **72.0 PSI**
(Measured in pounds per square inch)

Residual Pressure: **60.0 PSI**
(Measured in pounds per square inch)

Approx. distance between hydrants: 180 Feet

Main size: Not Provided

Flowing GPM: **2,946 GPM**

GPM @ 20 PSI: **6,503 GPM**

Scottsdale requires a maximum Static Pressure of 72 PSI for AFES Design.

Flow Test Location

North ↑

